# Mrittunjay Dubey

mrittunjaydubey3@gmail.com · LinkedIn · GitHub · Hugging Face

B.Tech CSE student (Fintech Honors, 9.85 CGPA) who builds AI systems from raw data to working interfaces, and reports their limits. Focused on secure systems, AI for healthcare and secure AI in fintech.

## Education

**SRM Institute of Science and Technology** | B.Tech CSE, Fintech Honors | Class of 2028 | CGPA 9.85

**CBSE** | Class 12: 92.2% | Class 10: 97.2%

## Projects

**APALA v2: seizure forecasting benchmark** | Python, PyTorch

- Built a leakage-controlled benchmark using nested leave-one-subject-out validation across 23 subjects and 624 fold trainings.
- Found no tested model beat a matched random alarm on CHB-MIT (13.2% vs 9.3% sensitivity, p = 0.185). Preprint, not peer reviewed.
- Showed how evaluation shortcuts turn 0 of 39 predicted seizures into 39 of 39 on an external dataset.

**MONA: language model built from scratch** | Python, PyTorch

- Trained a 110M-parameter transformer on 2.5B tokens of English, Hindi, Hinglish and Awadhi using one laptop GPU.
- Built an installable terminal app that teaches how LLMs work using the model's real weights.

**VARUNA: extreme weather forecasting** | PyTorch, NetCDF, HDF5

- Built a 17.6M-parameter Hybrid TransUNet that forecasts cloudbursts, thunderstorms and flash floods 2–6 hours ahead.
- Fused 190 GB of reanalysis, satellite and terrain data, reaching over 80% recall per hazard on an 11-window holdout, with 8–81% precision.

**CHB-MIT Preprocessed EEG: public dataset** | Python, Numba

- Published 347,075 labelled EEG windows with 590 features each, downloaded 407 times as of November 2026.

**GateKeeper: two-phone alert app for my parents** | Kotlin, Android, Firebase

- Built geofence-triggered alerts with a looping alarm, one-tap confirmation, SMS fallback and QR pairing.

## Skills

- **AI and ML:** Python, PyTorch, transformers, U-Nets, XGBoost, LightGBM
- **Software:** FastAPI, Next.js, WebSockets, Firebase, Android (Kotlin), pytest
- **Currently learning:** Java backend, DSA in C++, OOP, Computer Networks, Operating Systems, DBMS